

SMESA calculation geometries

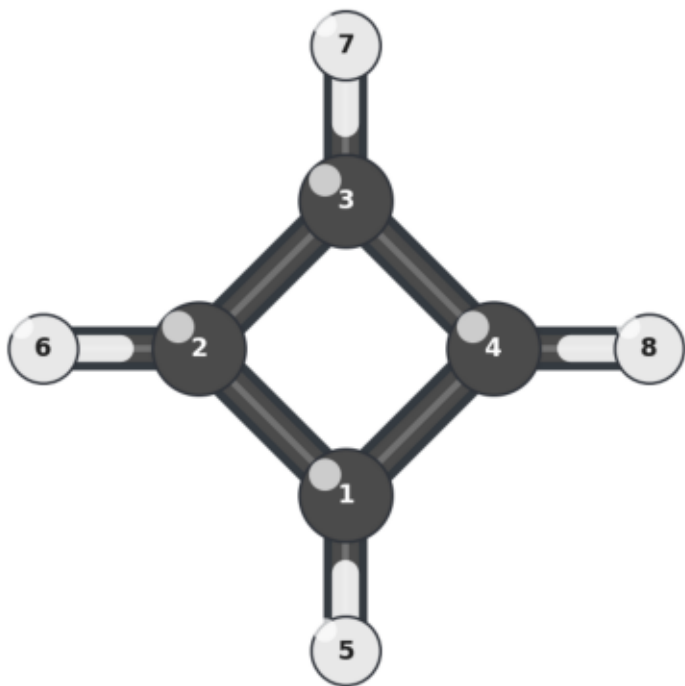
Supporting information for

Secondary Minimal Error Sampling Algorithms for Accelerating Self-Consistent Field Calculations

Paul Calistrate-Petre and Yan Alexander Wang

Cartesian coordinates and starting calculation settings

Coordinates in angstrom



Input	C4H4square/C4H4square.txt
Atoms	C4H4 (8 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF
Basis	LANL2DZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

#	Atom	x	y	z
1	C	1.018234	0.000000	0.000000
2	C	0.000000	1.018234	0.000000
3	C	-1.018234	0.000000	0.000000
4	C	0.000000	-1.018234	0.000000
5	H	2.097279	0.000000	0.000000
6	H	0.000000	2.097279	0.000000
7	H	-2.097279	0.000000	0.000000
8	H	0.000000	-2.097279	0.000000



Input	Co2/Co2.txt
Atoms	Co2 (2 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / UHF
Basis	LANL2DZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

#	Atom	x	y	z
1	Co	0.000000	0.000000	-1.100000
2	Co	0.000000	0.000000	1.100000



Input	Cr2/Cr2.txt
Atoms	Cr2 (2 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF
Basis	LANL2DZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

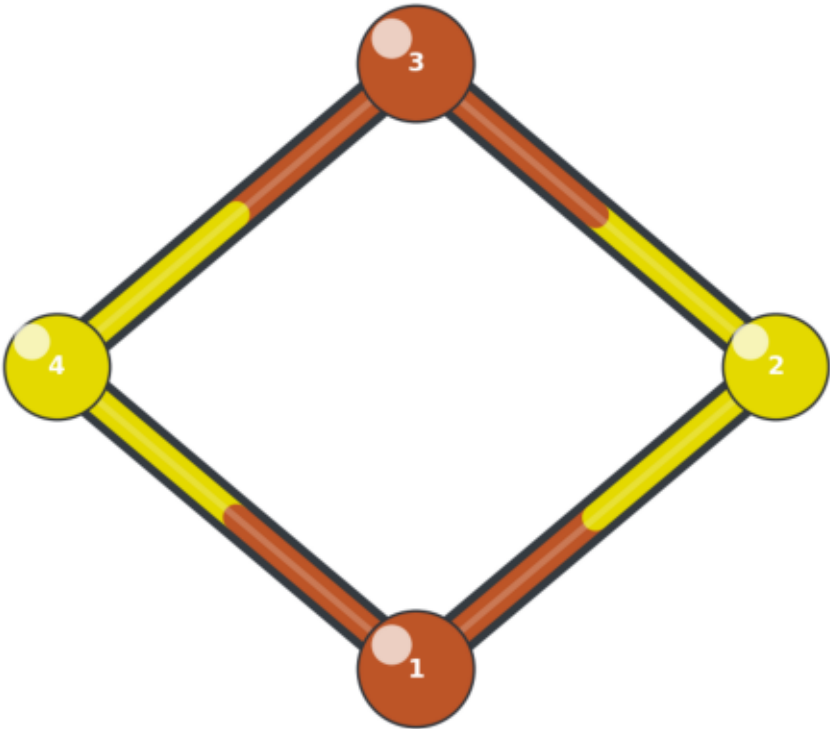
#	Atom	x	y	z
1	Cr	0.000000	0.000000	-1.680000
2	Cr	0.000000	0.000000	1.680000



Input	Cr3_linear/Cr3_linear.txt
Atoms	Cr3 (3 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / UHF
Basis	LANL2DZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

#	Atom	x	y	z
1	Cr	0.000000	0.000000	-2.000000
2	Cr	0.000000	0.000000	0.000000
3	Cr	0.000000	0.000000	2.000000



Input	Fe2S2/Fe2S2.txt
Atoms	Fe2S2 (4 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF
Basis	LANL2DZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

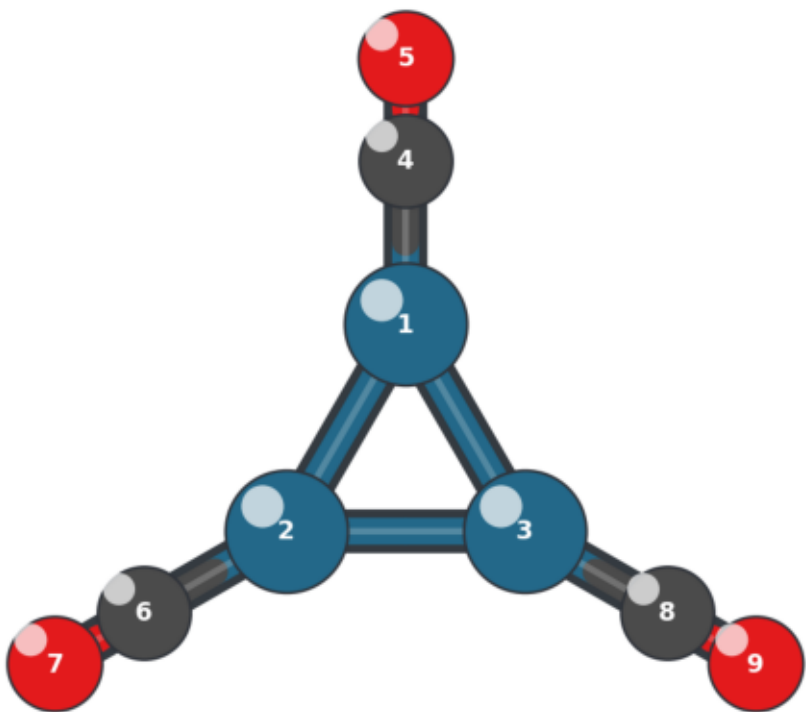
#	Atom	x	y	z
1	Fe	-0.000000	1.340534	0.000000
2	S	1.588225	0.000000	0.000000
3	Fe	0.000000	-1.340534	0.000000
4	S	-1.588225	-0.000000	0.000000



Input	Mn2_stretched/Mn2_stretched.txt
Atoms	Mn2 (2 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / UHF
Basis	LANL2DZ
Initial guess	SAD
SCF type	PK (default)

Cartesian geometry (angstrom)

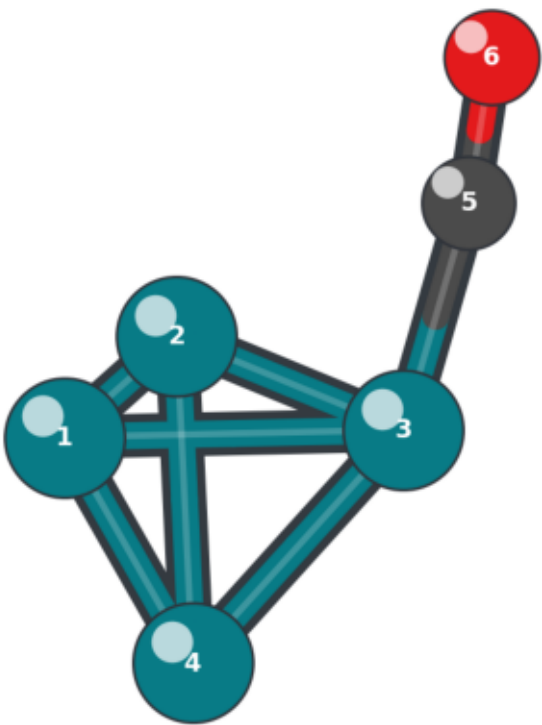
#	Atom	x	y	z
1	Mn	0.000000	0.000000	-1.575000
2	Mn	0.000000	0.000000	1.575000



Input	Os3C09/Os3C09.txt
Atoms	Os3C3O3 (9 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF
Basis	LANL2DZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

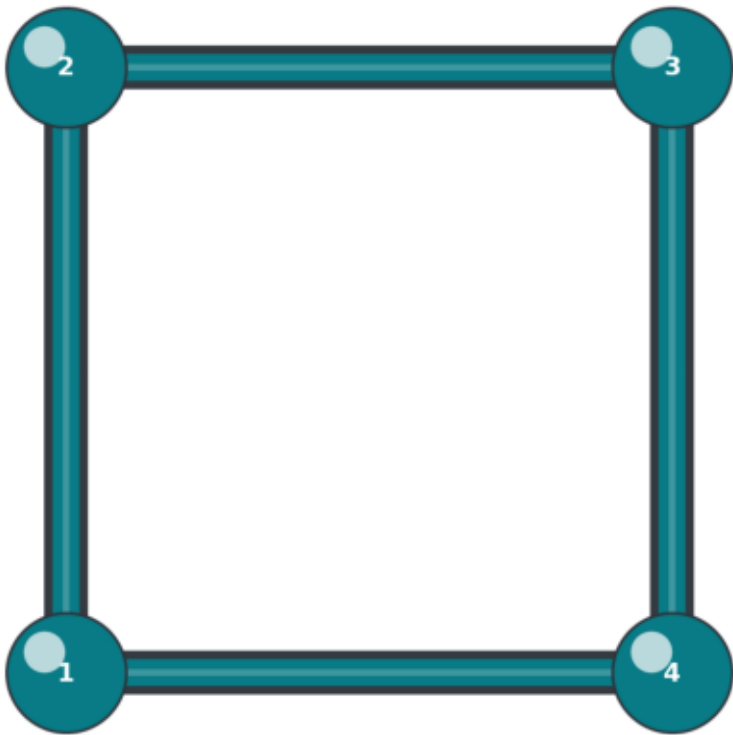
#	Atom	x	y	z
1	Os	0.000000	1.550000	0.000000
2	Os	1.342590	-0.775000	0.000000
3	Os	-1.342590	-0.775000	0.000000
4	C	0.000000	3.400000	0.000000
5	O	0.000000	4.550000	0.000000
6	C	2.947180	-1.700000	0.000000
7	O	3.943180	-2.275000	0.000000
8	C	-2.947180	-1.700000	0.000000
9	O	-3.943180	-2.275000	0.000000



Input	Rh4LCO/Rh4LCO.txt
Atoms	Rh4CO (6 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF (default)
Basis	LANL2DZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

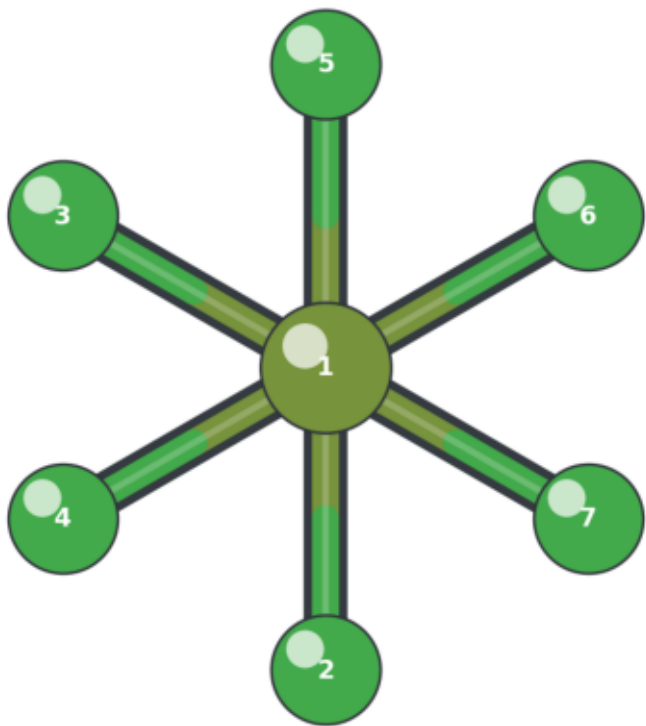
#	Atom	x	y	z
1	Rh	-0.825095	0.355937	-1.246735
2	Rh	-0.823385	0.358847	1.246145
3	Rh	1.373215	0.263967	-0.001715
4	Rh	-0.263835	-1.722033	0.001785
5	C	1.881075	2.076177	-0.000935
6	O	2.057115	3.224367	0.004055



Input	Rh4_square/Rh4_square.txt
Atoms	Rh4 (4 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF
Basis	LANL2DZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

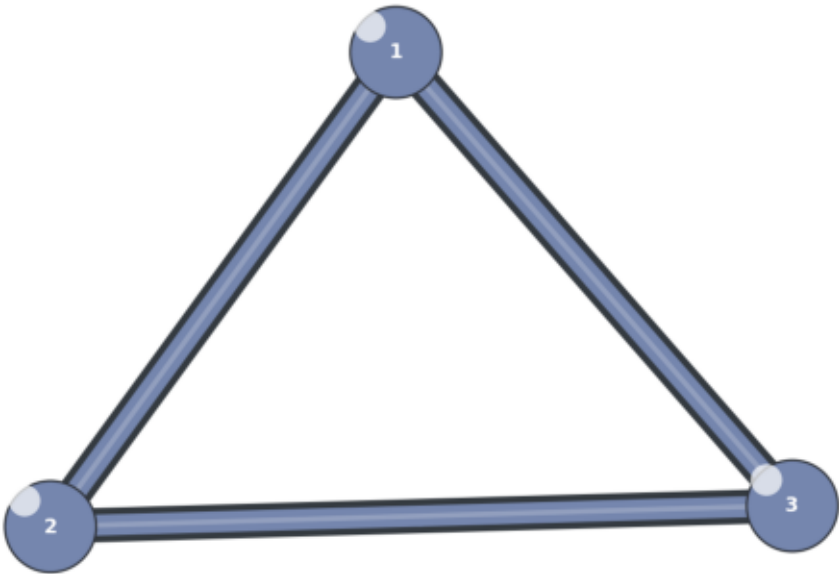
#	Atom	x	y	z
1	Rh	1.350000	1.350000	0.000000
2	Rh	-1.350000	1.350000	0.000000
3	Rh	-1.350000	-1.350000	0.000000
4	Rh	1.350000	-1.350000	0.000000



Input	UCI6/UCI6.txt
Atoms	UCI6 (7 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF (default)
Basis	LANL2DZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

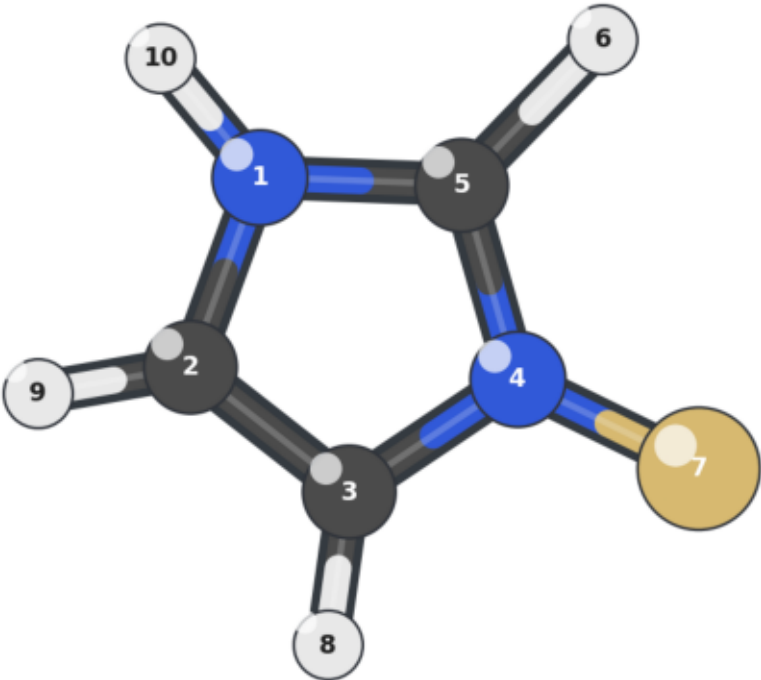
#	Atom	x	y	z
1	U	0.000000	0.000000	0.000000
2	Cl	0.000000	0.000000	2.390000
3	Cl	0.000000	2.390000	0.000000
4	Cl	-2.390000	0.000000	0.000000
5	Cl	0.000000	0.000000	-2.390000
6	Cl	2.390000	0.000000	0.000000
7	Cl	0.000000	-2.390000	0.000000



Input	V3_Scalene/V3_Scalene.txt
Atoms	V3 (3 atoms)
Charge / multiplicity	0 / 4
Method / reference	HF / UHF
Basis	LANL2DZ
Initial guess	GWH
SCF type	PK (default)

Cartesian geometry (angstrom)

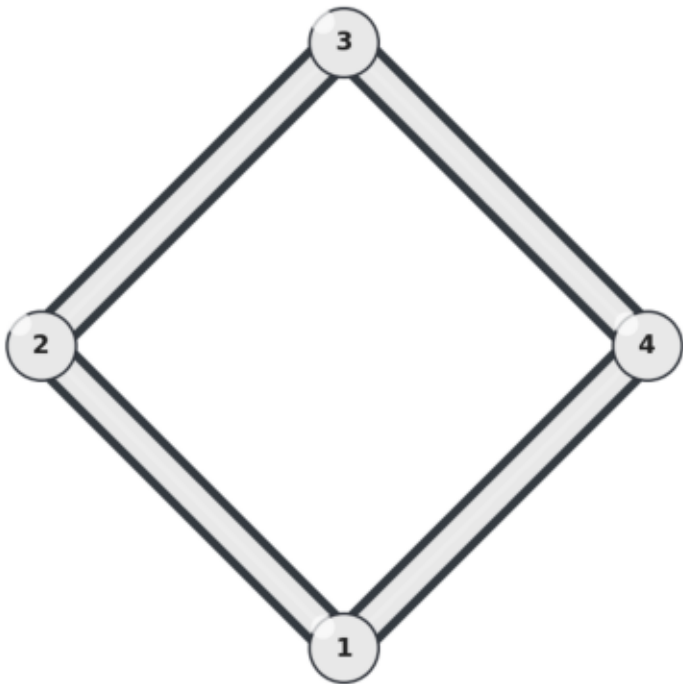
#	Atom	x	y	z
1	V	-0.900000	-0.733333	0.000000
2	V	1.300000	-0.733333	0.000000
3	V	-0.400000	1.466667	0.000000



Input	cdim/cdim.txt
Atoms	N2C3H4Cd (10 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF (default)
Basis	3-21G
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

#	Atom	x	y	z
1	N	-2.615701	0.796325	0.000000
2	C	-2.615701	-0.628175	0.000000
3	C	-1.260921	-1.068370	0.000000
4	N	-0.423621	0.084075	0.000000
5	C	-1.260921	1.236519	0.000000
6	H	-0.681525	2.537865	0.000000
7	Cd	0.993076	-0.064826	0.000000
8	H	-1.034297	-2.134551	0.000000
9	H	-3.559669	-1.173175	0.000000
10	H	-3.559669	1.341325	0.000000



Input	square_H4/square_H4.txt
Atoms	H4 (4 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF (default)
Basis	cc-pVQZ
Initial guess	GWH
SCF type	PK (default)

Cartesian geometry (angstrom)

#	Atom	x	y	z
1	H	1.414214	0.000000	0.000000
2	H	0.000000	1.414214	0.000000
3	H	-1.414214	0.000000	0.000000
4	H	0.000000	-1.414214	0.000000



Input	stretched_N2/stretched_N2.txt
Atoms	N2 (2 atoms)
Charge / multiplicity	0 / 1
Method / reference	HF / RHF (default)
Basis	cc-pVTZ
Initial guess	CORE (default)
SCF type	PK (default)

Cartesian geometry (angstrom)

#	Atom	x	y	z
1	N	0.000000	0.000000	-4.500000
2	N	0.000000	0.000000	4.500000